

BE 506: Biological Modelling and Simulation

Course Code	BE 506
Course Name	Biological Modelling and Simulation
L-T-P-C	2-0-2-3
Prerequisites	Understanding Biotechnology and its applications (IC136), and Bioinformatics (BE304), Computational Biology (BE505), or with permission of the instructor.
Intended for	Core for Integrated Dual Degree Bioengineering students with Specialisation in Computational Bioengineering, elective for other B.Tech and Mtech students, Elective for other UG, PG
Mutual Exclusion	None
Approval	43rd BoA (Oct 2021)

Course Contents

- **A. Modelling biomolecular structure, interactions and dynamics using atomistic simulations:** (Dr. D. Mohanty, NII) Introduction to Molecular Modelling, Molecular Mechanics (MM) Forcefields & empirical energy functions, Potential energy surface & Energy minimization algorithms, Molecular dynamics (MD), Explicit solvent simulations & Water models, Calculation of energy (E), temperature (T), Pressure (P) and Volume (V). Temperature control by velocity scaling and coupling to heat bath, Equilibration vs Production Dynamics, MD at higher temperatures for enhanced sampling & Simulated Annealing, Analysis of MD trajectories by Principal Component Analysis (PCA) and essential dynamics, MD Simulations in membrane environment, Calculation of Free Energy changes from explicit solvent MD by Free Energy Perturbation (FEP) approach, Monte Carlo (MC) Simulations, Atomistic vs coarse-grained dynamics. (13 hours)
- **B. Systems Biology & Biological Networks:** (Prof James Gomes, IIT Delhi) (2 hours)
- **C. Simulation of cellular subsystems:** Simulation & Analysis of Biochemical Network Models (3 Hours) (Prof James Gomes, IIT Delhi) Simulation of Genome-Scale Metabolic (GSM) Networks, Flux-Balance Analysis (FBA) & Constraint based models (5 Hours) (Dr. Karthik Raman, IIT Chennai) Introduction to Boolean Network Modeling of Gene Regulation (2 hours) (Dr D. Mohanty, NII). (11 hours)
- **D. Population models:** (Dr. Tulika P Srivastava, IIT Mandi) (3 hours)

Lab Course content: (28 hours)

The below mentioned 10 topics will be covered over the 14 weeks:

- **A. Atomistic Simulations** (12 hours) (Dr. D. Mohanty & Tulika P. Srivastava)
 - Visualization and analysis of 3D structures of biomolecules and Model building.
 - Energy minimization.
 - Molecular Dynamics Simulation of a Protein using GROMACS.
 - Analysis of MD trajectory & Principal Component Analysis (PCA) of MD trajectory.
 - Calculation of Free Energy (solvation of methane or amino acids in water) using explicit water simulations.
 - MD simulations of a Protein-Ligand complex.
 - MD simulation of a model transmembrane peptides in lipid bilayers.
- **B. Analysis of biological networks** (3 hours)
 - Visualization of biological networks and calculation of network parameters using Cytoscape.
- **C. Simulation of cellular subsystems** (8 hours)
 - Simulation & Analysis of biochemical network models using differential equations (COPASI).
 - Flux Balance Analysis using COBRA Tool for simulation of genome scale metabolic networks: Applications to central metabolism of E. coli.
- **D. Population models** (5 hours)
 - Predator-Prey Simulation
 - Modeling spread of infectious disease: COVID19.

Text Books

1. Andrew R. Leach, **Molecular Modelling: Principles and Applications**, 2nd Edition.
2. Karthik Raman, **An Introduction to Computational Systems Biology Systems-Level Modelling of Cellular Networks**, Chapman and Hall/CRC, 2021.

References

1. JA McCammon & SC Harvey, **Dynamics of Proteins & Nucleic Acids**, Cambridge University Press.